Adrien Carrou

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Work Experience

Staff Software Development Engineer - Intuitive Machines **Dec 2023 – Present**

- Develop and maintain C/C++ flight software for mechanism control, timing, radio link and sensor interfaces on a cFS/FreeRTOS flight computer.
- Serve as flight software point of contact for payload and customer integration requests, turning them into Jira tickets, driving them to closure, and delivering formal verification reports that closed nine customer requirements in 2026.
- Plan and run test campaigns across simulation and engineering-model hardware benches, coordinating test, simulation, GNC and mechanical engineers, and writing the procedures and runbooks other engineers execute from.
- Triage defects from first report to merged fix across a multi-repository flight software baseline, coordinating reviews, submodule pins and release tagging with a distributed team.
- Received the President's Spotlight Award as part of a team recognized for exceptional collaboration, problem-solving, and delivery.

Founder and Tech Lead - Nexus Analytica https://www.nexusanalytica.net/ **June 2024 – Present**

- Founded a company specializing in multispectral and hyperspectral imaging systems.
- Lead embedded firmware and hardware development, integrating sensors and building data acquisition pipelines.
- Develop custom firmware, multilayer PCBs, and FPGA-based acquisition logic to meet client-specific reliability and performance requirements.

Test Automation Engineer - Maxar Technologies **May 2022 – Dec 2023**

- Built simulation and test-automation pipelines for satellite flight software verification, including configuration management and reproducible-run procedures.
- Wrote Python tooling for log parsing, test harness setup, and result aggregation across simulated and real-hardware runs.
- Authored configuration, runbook, and integration documentation for downstream test engineers.

Leadership Experience

Club Leadership - SJSU Robotics and Cube3 **Sep 2021 – Aug 2023**

- Led multidisciplinary teams of 11–15 students across robotics and small-satellite projects, guiding technical development and subsystem integration through launch preparation.

Education

Johns Hopkins University (JHU) M.S. in Electrical and Computer Engineering

San Jose State University (SJSU) B.S. in Computer Engineering

Skills

Development & Collaboration Tools: Jira, Git, GitLab, merge request workflows, Docker, CMake

Programming: C/C++, Python, Bash, Tcl, MATLAB, VHDL, Verilog

Systems & Testing: Flight software, embedded systems, cFE/cFS, FreeRTOS, Linux, Google Test, hardware-in-the-loop and software-in-the-loop testing

Process: Agile/Scrum, Jira-driven ticket workflows, GitLab merge requests and CI, requirements verification and test reporting, cross-team engineering coordination